LEVELHULL · PROJECT PLAN

LevelHull project plan

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	LVH-PLN-001	Version	0.2
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
Source	docs/11-plan.md @ e0fd582		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	D-O2, D-O3 and D-R1 decided by Amish (2026-10-25); freeing-port study at TRL 4
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade)

LevelHull project plan

LevelHull is at TRL 3 and stage P3 (design outputs): model at D3 for all three concept options, drawings, priced BOM, build plan and the pipeline documents exist. The next gate is P4 (verification), which is on hold under the portfolio cap; the decisions requested of Amish now are the concept choice between options A and B (LVH-DEC-001 D-O1), which scored within 10 % of each other. D-O2, D-O3 and D-R1 were decided by Amish on 2026-10-25. Method: STANDARDS sections 19 and 23.

> **Safety:** P4 work (pull tests, swamp tests) only starts on Amish's go and only under the safety stops of LVH-BLD-001 section 6: sheltered shallow water, a safety boat and every person in a life jacket.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept choice	Amish	Decision D-O1 (A or B)	Awaiting Amish
Design and model	Design lead	`cad/src/model.py`, D3 for A, B and C	Done (P3)
Calculations and STEM check	Design lead	LVH-CAL-001, LVH-STM-001	Done; three field items open
Requirements and traceability	Design lead	LVH-URS-001, LVH-PRS-001, LVH-RTM-001	Done; PR-014 to PR-017 accepted by Amish, 2026-10-25
Build plan	Design lead	LVH-BLD-001	Done (plan, not yet built)
Partner and approvals	Project lead	A partner BMU; the KMA question (LVH-REG-001 G1)	Not started
Renders and storefront	Coordinator	Options picture, product renders, cards (on Amish's Mac)	To do

Resources needed

Table 2. Resources (P4 and later on hold).

Resource	Deliverable	Effort	Estimated cost (USD)
Co-design partner (BMU and canoe builders)	Hull and load survey; first fitting	5 to 10 days	To assess
Kit parts for two canoes	Prototype kits		1,360 (BOM; value-engineering target 1,000)
CalRig proof-load rig	Fixing pull and proof tests (P4)	2 days	To assess
Safety boat and crew	Swamp tests (P4)	2 days	To assess
Regulatory adviser	KMA approval question	1 to 2 days	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Done (2026-10-24)
P1 Definition	URS, PRS draft, RSK with hazards, MKT	Done (2026-10-24)
P2 Design inputs	PRS with targets, CAL, RTM, concept selection DDR, STEM check closed where calculation can close it	Done; concept choice awaiting Amish (LVH-DDR-004)
P3 Design outputs	Model, STEP, drawings, priced BOM, BLD, DEC, design FMEA, traceability	Done (2026-10-24)
P4 Verification	Pull tests, proof load, swamp tests, immersion of foam samples (TST)	On hold (cap P3)
P5 Validation	Field trial with a partner fishery, durability over 12 months (PR-009)	On hold
P6 Commercialization	Builder training pack, only with a named deployment partner	On hold

Next steps and decision requested

1. **Amish:** choose between option A (the design) and option B (external sponsons), or keep A and consider lowering its modules by about 40 mm (B's lesson); LVH-DEC-001 D-O1, LVH-DDR-004.
2. Done 2026-10-25: R2 and R3 stated for the free-flooding condition with the operating rule (D-O2; freeing ports studied at TRL 4 with the partner), four bailers (D-O3), PR-014 to PR-017 accepted (D-R1).
3. **TRL 4, with the partner:** study one-way freeing ports (D-O2 (a)) and time the bail-out from brim-full with four bailers.
4. Coordinator: render the options picture and product renders from the exported scenes.
5. When the cap allows: approach the first partner candidate and put the KMA question.

References

- FAO, ILO and IMO. *Safety Recommendations for Decked Fishing Vessels of Less than 12 metres in Length and Undecked Fishing Vessels*. Rome: FAO, 2012. https://www.fao.org/4/i3108e/i3108e00.htm. Accessed October 25, 2026.
- Google Patents. "US9139267B2." https://patents.google.com/patent/US9139267B2/en.
- ISO. *ISO 12217-1:2022, Small craft: Stability and buoyancy assessment and categorization, Part 1*. Scope at https://www.evs.ee/en/iso-12217-1-2022. Accessed October 25, 2026.
- KB Packaging. "Foam Sheet 2000mm x 1200mm x 50mm." https://kbpackaging.com/product/foam-sheet-2000mm-x-1200mm-x-50mm/. Accessed October 25, 2026.
- Kenya. *The Merchant Shipping (Small Vessel Safety) Regulations*. Legal Notice 114 of 2012. https://new.kenyalaw.org/akn/ke/act/ln/2012/114/eng@2022-12-31. Accessed October 25, 2026.
- Ravago Building Solutions. "Ethafoam Technical Data Sheet." https://ravagobuildingsolutions.com/gr/wp-content/uploads/sites/47/2022/01/tds-ethafoam-eng.pdf. Accessed October 25, 2026.
- RNLI. "Sustainable Lifesaving Equipment." https://rnli.org/what-we-do/international/lifesaving-interventions/sustainable-lifesaving-equipment.
- The Conversation. "Lake Victoria: Why So Many Fishers Are Dying and What Can Be Done about It." https://theconversation.com/lake-victoria-why-so-many-fishers-are-dying-and-what-can-be-done-about-it-232006.
- US Coast Guard. *Safe Boating: Flotation* (33 CFR 183.220 to 183.235). https://www.uscgboating.org/assets/PDF/downloads/FLOTATION.pdf. Accessed October 25, 2026.
- Winter, David A. *Biomechanics and Motor Control of Human Movement*. 4th ed. Hoboken: Wiley, 2009.
- Zoro. "Onyx Life Jacket, Adult, Universal, 15.5 lb, Foam." https://www.zoro.com/i/G305778328/. Accessed October 25, 2026.
- CEN. EN 1995-1-1, *Eurocode 5: Design of Timber Structures, Part 1-1*; EN 338, *Structural Timber: Strength Classes*.